

Global Superstore

Sales Analytics — Data Project Report

SQL Database Design · Power BI Dashboard · Analytical Insights

May 2026

51,290+
Order Records

10
SQL Queries

5
Dim Tables

1
Fact Table

5
Dashboard Pages

1. Dataset Description

The **Global Superstore** dataset is a retail analytics dataset covering orders across multiple global markets. The source table **superstore_complete_2** contains 30 fields spanning customer, product, location, shipping, and financial dimensions. Five additional engineered features were added before loading into the star-schema warehouse.

Field Group	Key Fields	Purpose
Order Identity	order_id, order_date, ship_date	Track each transaction and fulfilment timeline
Customer	customer_id, customer_name, segment, market	Identify customers and business segments
Geography	country, state, city, region	Multi-level spatial hierarchy
Product	category, sub_category, product_name, product_id	Three-level product taxonomy
Financials	sales, quantity, discount, profit, shipping_cost	Core transaction KPIs
Derived Metrics	profit_margin, sales_per_unit	Calculated financial ratios
Engineered – Time	order_year/month/quarter/week/day_of_week	Date-part extraction for trend analysis
Engineered – Flags	is_weekend, discount_bin	Behavioural flag and discount tier

Table 1 — Source dataset fields grouped by category

1.1 Engineered Features

- **profit_margin** = profit ÷ sales — reveals true profitability independent of order volume.
- **sales_per_unit** = sales ÷ quantity — normalises revenue across varying order sizes.
- **discount_bin** — categorical bucketing: 0 %, 1–20 %, 21–40 %, 40 %+ for segmented analysis.
- **is_weekend** — binary flag from order_day_of_week for weekday vs. weekend behaviour.
- **order_year / month / quarter / week** — extracted from order_date for time-hierarchy slicing.

2. SQL Queries — Explained

Ten analytical queries were written against the star schema using JOIN, GROUP BY, HAVING, and subquery patterns to answer distinct business questions.

Q1 — Total Sales & Profit per Year

Aggregates SUM(sales) and SUM(profit) grouped by year via a JOIN to dim_date. Reveals revenue trends and whether profitability kept pace with growth.

```
SELECT d.order_year, SUM(f.sales), SUM(f.profit)
FROM fact_sales f
JOIN dim_date d ON f.order_date = d.date_key
GROUP BY d.order_year ORDER BY d.order_year;
```

Q2 — Top 5 Products by Sales

Uses TOP 5 ... ORDER BY DESC to identify the highest-revenue SKUs instantly.

```
SELECT TOP 5 p.product_name, SUM(f.sales) AS total_sales
FROM fact_sales f
JOIN dim_products p ON f.product_key = p.product_key
GROUP BY p.product_name ORDER BY total_sales DESC;
```

Q3 — High-Value Customers (HAVING)

HAVING SUM(sales) > 5000 filters only customers with meaningful revenue — avoids row-level WHERE.

```
SELECT c.customer_name, SUM(f.sales)
FROM fact_sales f
JOIN dim_customers c ON f.customer_id = c.customer_id
GROUP BY c.customer_name
HAVING SUM(f.sales) > 5000;
```

Q4 — Order Count per Region

COUNT(DISTINCT order_id) measures true order volume, excluding duplicate line-item rows.

```
SELECT l.region, COUNT(DISTINCT f.order_id) AS total_orders
FROM fact_sales f
JOIN dim_location l ON f.location_key = l.location_key
GROUP BY l.region;
```

Q5 — Products Above Average Sales

Scalar subquery computes global AVG(sales); outer HAVING filters products exceeding that threshold.

```
HAVING SUM(f.sales) > (SELECT AVG(sales) FROM fact_sales)
```

2. SQL Queries — Continued

Q6 — Profit Margin per Segment

SUM(profit)/SUM(sales) correctly weights the margin across Consumer, Corporate & Home Office.

```
SELECT c.segment, SUM(f.profit)/SUM(f.sales) AS profit_margin
FROM fact_sales f
JOIN dim_customers c ON f.customer_id = c.customer_id
GROUP BY c.segment;
```

Q7 — Monthly Sales Trend

Two-column GROUP BY (year, month) produces a time series for seasonality and YoY comparison.

```
GROUP BY d.order_year, d.order_month
ORDER BY d.order_year, d.order_month;
```

Q8 — Highest Single Sales Order

Scalar subquery MAX(sales) locates the peak transaction in one pass — useful for anomaly detection.

```
SELECT * FROM fact_sales
WHERE sales = (SELECT MAX(sales) FROM fact_sales);
```

Q9 — Top Shipping Modes by Profit

Reveals whether express shipping erodes margins vs. standard modes — key for logistics optimisation.

```
SELECT s.ship_mode, SUM(f.profit) AS total_profit
FROM fact_sales f
JOIN dim_shipping s ON f.ship_key = s.ship_key
GROUP BY s.ship_mode ORDER BY total_profit DESC;
```

Q10 — vw_sales_summary View

Creates a reusable view (year × segment × category) to simplify Power BI refresh and ad-hoc queries.

```
CREATE VIEW vw_sales_summary AS
SELECT d.order_year, c.segment, p.category,
       SUM(f.sales), SUM(f.profit)
FROM fact_sales f
JOIN dim_date d ... JOIN dim_customers c ...
GROUP BY d.order_year, c.segment, p.category;
```

3. Power BI Dashboard Insights

The Power BI file connects to the star schema via five analytical pages. Each page targets a distinct business question using slicers, KPI cards, and charts.

Dashboard Page	Key Insights
Sales Overview	APAC and US lead in volume. KPI cards show total sales, profit and average discount at a glance. Year-over-year trend confirms consistent ~20 % revenue growth from 2012 to 2015.
Product Performance	Technology is the dominant category by both revenue and margin. Copier and Phone models top the Q2 ranking. A scatter plot exposes high-revenue / low-margin products needing pricing attention.
Customer Analysis	Consumer segment holds 51 % share; Corporate 30 %; Home Office 19 %. High-value customers (Q3) feed a CRM watchlist. LATAM appears under-monetised relative to order count.
Shipping & Logistics	Standard Class carries the best absolute profit. Same Day shipping shows the lowest margin — evidence of cost erosion. Critical orders are over-represented in high-cost ship modes.
Time Intelligence	Q4 spikes signal seasonal demand concentration requiring proactive stock planning. Weekday orders generate 87 % of revenue. Monthly trend (Q7) enables granular seasonality analysis.

Table 2 — Power BI dashboard pages and key findings